Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division
First / Second Semester 20-20****

Mid-Semester Test
(EC-2 Regular/ EC-2 Make-up)

Course No.	: ****
Course Title	: ****
Nature of Exam	: Closed Book / Open Book (As per Course Handout)
Weightage	: 30% or 35% (As per Course Handout)
Duration	: 2 Hours
Date of Exam	: **** (FN/AN)

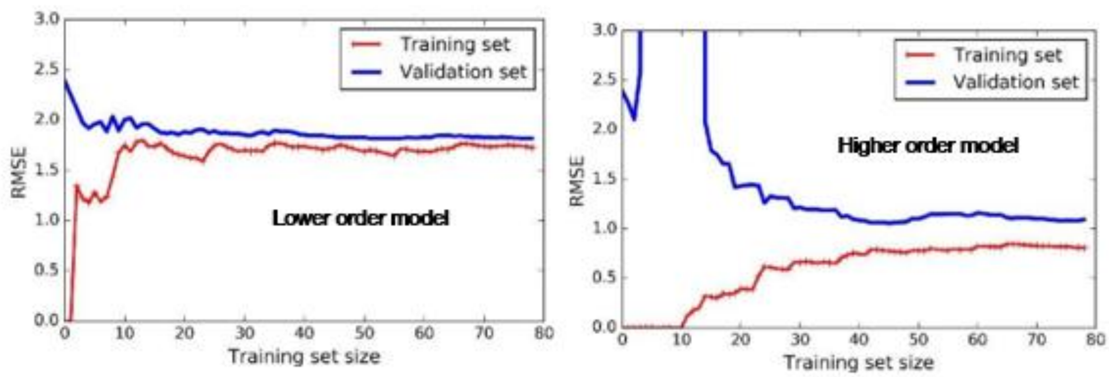
No. of Pages =
No. of Questions =

Note to Students:
1. Please follow all the *Instructions to Candidates* given on the cover page of the answer book.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
3. Assumptions made if any, should be stated clearly at the beginning of your answer.

Please Note:
A **maximum of 12 main questions** with Q.Nos. 1 to N**.**

Q.1. [3 Marks]

You are training two polynomial regression models (Degree = 3 and Degree = 10) on a dataset of **80 samples** to predict housing prices.
The training and validation RMSE trends with dataset size are shown below (qualitatively similar to the slide).



After observing the curves, your manager suggests:

“Let’s always choose the higher-order model since it gives lower training error.”

Question:

a) Evaluate this suggestion (0.5 marks)
b) Explain what could go wrong when deploying the model on unseen data. (1 mark)
c) Then propose **three ways** to ensure the higher-order model generalizes better (1.5 marks)

Solution

a) High-order model may **memorize** training data → low training error but high validation error. (0.5 marks)

b) On unseen data → poor generalization (Overfitting). (1 mark)
c) Solutions may include **regularization (Ridge/Lasso)**, **cross-validation**, **early stopping**, **simplifying model**, or **data augmentation**. (1.5 marks)

Q.2. [6 Marks]

A company wants to predict whether a customer will repay a loan. The training dataset contains 14 samples, out of which 9 are labeled "Repay" and 5 are labeled "Default."
The attribute Income Level has two values: High and Low. The dataset splits as follows:
- Income = High: 6 Repay, 2 Default
- Income = Low: 3 Repay, 3 Default

Using entropy-based information gain, determine whether Income Level is a good attribute to split on. Show all steps.

Solution

1. Entropy of parent set:
$H(S) = - (9/14) \log2(9/14) - (5/14) \log2(5/14) \approx 0.94$

2. Entropy of child nodes:
Income = High:
$H = - 6/8 \log2(6/8) - 2/8 \log2(2/8) \approx 0.81$

Income = Low:
$H = - 3/6 \log2(3/6) - 3/6 \log2(3/6) = 1.0$

3. Weighted entropy after splitting:
$H\_after = (8/14)(0.81) + (6/14)(1.0) \approx 0.8915$

4. Information Gain:
$Gain = H(S) - H\_after = 0.94 - 0.8915 \approx 0.0485$

Conclusion: Low information gain (≈0.05). Income Level is not a good attribute for splitting.

Rubric (6 Marks):
- Parent entropy calculation: 1.5 marks
- Child entropy calculations: 2 marks
- Weighted entropy calculation: 1 mark
- Information gain calculation: 1 mark
- Interpretation: 0.5 mark

Q.3. [3+3+2 = 8 Marks]

Consider the following dataset:

x	1	2	3
y	2	5	7

We assume a simple linear regression model

$$h_\theta(x) = \theta_0 + \theta_1 x.$$

The regularized cost function is defined as

$$J(\theta_0, \theta_1) = \frac{1}{2m}\sum_{i=1}^{m}\left(h_\theta(x_i) - y_i\right)^2 + \frac{\lambda}{2m}\theta_1^2,$$

where

$$m = 3, \qquad \lambda = 2.$$

You are given:

$$\theta_0^{(0)} = 0, \qquad \theta_1^{(0)} = 1, \qquad \alpha = 0.05, \qquad m = 3.$$

Answer the following:

1. Write the expressions for the hypothesis $h_\theta(x)$ and the regularized cost function $J(\theta_0, \theta_1)$ for this dataset. Derive the gradient descent update rules for θ_0 and θ_1 for this regularized cost function, that is,

$$\theta_0 := \theta_0 - \alpha\frac{\partial J}{\partial \theta_0}, \qquad \theta_1 := \theta_1 - \alpha\frac{\partial J}{\partial \theta_1}.$$

2. Using the initial values $\theta_0^{(0)} = 0$ and $\theta_1^{(0)} = 1$, perform **one full iteration of batch gradient descent** and compute the updated parameters

$$\theta_0^{(1)}, \qquad \theta_1^{(1)}.$$

3. Compute the cost $J(\theta_0^{(0)}, \theta_1^{(0)})$ and the cost $J(\theta_0^{(1)}, \theta_1^{(1)})$. Comment on whether the cost has decreased after this iteration.

Solution:

(1) Hypothesis, Cost, Gradients and Update Rules (3 Marks)

$$h_\theta(x) = \theta_0 + \theta_1 x,$$

$$J(\theta_0, \theta_1) = \frac{1}{2m}\sum_{i=1}^{3}\left(\theta_0 + \theta_1 x_i - y_i\right)^2 + \frac{\lambda}{2m}\theta_1^2.$$

Gradients and Update Rules

The partial derivatives are

$$\frac{\partial J}{\partial \theta_0} = \frac{1}{m}\sum_{i=1}^{m}\left(h_\theta(x_i) - y_i\right),$$

$$\frac{\partial J}{\partial \theta_1} = \frac{1}{m}\sum_{i=1}^{m}\left(h_\theta(x_i) - y_i\right)x_i + \frac{\lambda}{m}\theta_1.$$

Thus the update rules are:

$$\theta_0 := \theta_0 - \alpha\cdot\frac{1}{m}\sum_{i=1}^{m}\left(h_\theta(x_i) - y_i\right),$$

$$\theta_1 := \theta_1 - \alpha\left[\frac{1}{m}\sum_{i=1}^{m}\left(h_\theta(x_i) - y_i\right)x_i + \frac{\lambda}{m}\theta_1\right].$$

(2) One Iteration from $\theta_0^{(0)} = 0,\ \theta_1^{(0)} = 1$ (3 Marks)

Data points:

$$(x_1, y_1) = (1, 2), \quad (x_2, y_2) = (2, 5), \quad (x_3, y_3) = (3, 7), \quad m = 3.$$

At iteration 0:

$$\begin{aligned} h_\theta(x_1) &= 0 + 1 \cdot 1 = 1, \quad e_1 = 1 - 2 = -1, \\ h_\theta(x_2) &= 0 + 1 \cdot 2 = 2, \quad e_2 = 2 - 5 = -3, \\ h_\theta(x_3) &= 0 + 1 \cdot 3 = 3, \quad e_3 = 3 - 7 = -4. \end{aligned}$$

Then

$$\sum e_i = -1 - 3 - 4 = -8, \qquad \sum e_i x_i = (-1) \cdot 1 + (-3) \cdot 2 + (-4) \cdot 3 = -1 - 6 - 12 = -19.$$

Update for θ_0:

$$\theta_0^{(1)} = \theta_0^{(0)} - \alpha \cdot \frac{1}{m} \sum e_i = 0 - 0.05 \cdot \frac{1}{3}(-8) = 0.05 \cdot \frac{8}{3} = \frac{0.4}{3} \approx 0.1333.$$

Update for θ_1:

$$\frac{1}{m} \sum e_i x_i = \frac{-19}{3}, \qquad \frac{\lambda}{m} \theta_1^{(0)} = \frac{2}{3}.$$

$$\frac{1}{m} \sum e_i x_i + \frac{\lambda}{m} \theta_1^{(0)} = \frac{-19}{3} + \frac{2}{3} = \frac{-17}{3}.$$

$$\theta_1^{(1)} = 1 - 0.05 \left(\frac{-17}{3} \right) = 1 + 0.05 \cdot \frac{17}{3} = 1 + \frac{0.85}{3} \approx 1.2833.$$

So approximately:

$$\theta_0^{(1)} \approx 0.1333, \qquad \theta_1^{(1)} \approx 1.2833.$$

(3) Cost Before and After (2 Marks)

Initial squared errors:

$$e_1^2 = 1, \quad e_2^2 = 9, \quad e_3^2 = 16, \quad \sum e_i^2 = 26.$$

Thus,

$$\frac{1}{2m} \sum e_i^2 = \frac{1}{6} \cdot 26 = \frac{26}{6}, \qquad \frac{\lambda}{2m} (\theta_1^{(0)})^2 = \frac{2}{6}.$$

$$J(\theta_0^{(0)}, \theta_1^{(0)}) = \frac{26}{6} + \frac{2}{6} = \frac{28}{6} \approx 4.6667.$$

Students can similarly compute $J(\theta_0^{(1)}, \theta_1^{(1)})$ and should observe that the cost decreases, indicating that gradient descent has moved the parameters in a direction that reduces the cost.

Q.4. [5 Marks]

A medical diagnostic system is tested on 1,000 patients, of whom 150 have the disease. The system correctly identifies 120 diseased patients and incorrectly flags 85 healthy patients as diseased.

(a) Construct the confusion matrix (1 mark)

(b) Calculate accuracy, precision, recall, and F1-score (2 marks)

(c) If the cost of a false negative is $50,000 (missed treatment) and a false positive is $2,000 (unnecessary tests), calculate the total cost of errors. What threshold adjustment would you recommend? (2 marks)

Solution:
(a) Confusion Matrix: (1 mark)

	Predicted Positive	Predicted Negative
Actual Positive	120 (TP)	30 (FN)
Actual Negative	85 (FP)	765 (TN)

(b) Metrics:
- Accuracy = (120 + 765)/1000 = 88.5% (not 95%!)
- Precision = 120/(120 + 85) = 120/205 = 58.5%
- Recall (Sensitivity) = 120/150 = 80%
- F1-Score = 2 × (0.585 × 0.80)/(0.585 + 0.80) = 0.936/1.385 = 67.6%

(0.5 mark each = 2 marks)
(c) Cost Analysis:
- Cost of FN: 30 × $50,000 = $1,500,000
- Cost of FP: 85 × $2,000 = $170,000
- Total cost: $1,670,000 (1 mark)

Recommendation: Lower the classification threshold to increase recall (catch more diseased patients). The FN cost is 25× higher than FP cost, so we should prioritize reducing false negatives even at the expense of more false positives. (1 mark)

Q.5. [2 Marks]
A model has learned the logistic regression parameters θ for a dataset with 4 features (Age, CGPA, IQ, Height). A student claims that a large positive θ_2 (weight of CGPA) "guarantees" that candidates with high CGPA are always selected.
Is this conclusion correct? Justify your answer using the log-odds interpretation of the conclusion.

Solution

No, this is not correct (0.5 marks)
<u>Justification on below lines (1.5 marks)</u>
Logistic regression models log-odds, not direct decisions.
A large positive θ_2 increases log-odds of selection, but overall probability still depends on:
θ_0 (bias term), other weights θ_1, θ_3 …, actual value ranges of other features,
Even with high CGPA, a low IQ or strong negative weight for another feature can shift $\theta^T x$ to a negative value → $h\theta(x) < 0.5$.
Therefore, selection is based on combined weighted features, not one feature alone.

Q.6. [6 Marks]

You are given an original dataset for predicting **whether a house sale will be "High Price" (1) or "Low Price" (0)** using two features:

- **area** (in m^2)
- **age** (in years)

Three models are trained on this original dataset:

- **Model L1:** Linear Regression (used for predicting a continuous price)
- **Model L2:** Logistic Regression (for High/Low classification)
- **Model T:** Decision Tree Classifier

Now we create **two modified datasets**:

Dataset A – Noisy / Irrelevant Features Added

To the original features (area, age), we add two completely random, irrelevant features:

- **noise1:** random number between 0 and 1
- **noise2:** random number between −1 and 1

These noise features have **no real relationship** with the target (High/Low price), but they are included while training all three models again.

Observation: Training performance improves (lower training error / higher training accuracy), but test performance drops.

Dataset B – Outliers Introduced

We keep only the original two features (area, age), but now we **introduce outliers**:

- A few houses with **extremely large area** (10× bigger than typical)
- A few records with **age = 0 or 100+ years**, far outside the normal range

The added points are still labelled correctly (High/Low), but they are **extreme** compared to the rest of the data.

All three models are trained again on this dataset.

Question:

For **each model (L1, L2, T)** and **each dataset (A: noisy features, B: outliers)**, indicate whether the model is **strongly affected** or **relatively less affected**, and give a short conceptual justification for your answer.

[Affected: clearly increases risk of overfitting / harms generalization
Less affected: comparatively more robust in that scenario (but not immune)]

You may summarise your answer in a table format like the one below, with brief justifications.

Table for Dataset A:

Model	Affected/ Less Affected	Short Justification
Linear Regression (L1)		
Logistic Regression (L2)		
Decision Tree (T)		

Table for Dataset B:

Model	Affected/ Less Affected	Short Justification
Linear Regression (L1)		
Logistic Regression (L2)		
Decision Tree (T)		

Solution:

Legend

- **Affected:** clearly increases risk of overfitting / harms generalisation
- **Less affected:** comparatively more robust in that scenario (but not immune)

(i) Dataset A – Noisy / Irrelevant Features Added

Model	Affected?	Short Justification
Linear Regression (L1)	**Affected**	Fits a weight for every feature; noisy features get non-zero weights due to spurious correlations, reducing training error but increasing variance and hurting test performance.
Logistic Regression (L2)	**Affected**	Also estimates weights for all features; noise can distort the decision boundary as the model tries to exploit random patterns → overfitting, lower test accuracy.
Decision Tree (T)	**Strongly affected**	Tree greedily splits on any feature that offers even small impurity reduction; noisy features can create extra branches that perfectly fit random quirks in training data → very high training accuracy but poor generalisation (high variance).

(ii) Dataset B – Outliers Introduced

Model	Affected?	Short Justification
Linear Regression (L1)	**Strongly affected**	Minimises squared error, so extreme outliers pull the regression line towards them, heavily distorting the fit and harming both training and test behaviour.
Logistic Regression (L2)	**Less affected (but not immune)**	Sigmoid squashes extreme values; probabilities saturate near 0 or 1, so single extreme points don’t dominate as much as in linear regression. Still, outliers can slightly shift the decision boundary and affect the fitted weights.
Decision Tree (T)	**Affected (moderately–strongly)**	Outliers can cause unusual splits (e.g., separate branch just for a few extreme points). Trees are less “pulled” globally than linear models, but locally they may overfit those rare extreme cases, increasing variance.

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